

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSECODE:ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage:50%

QUESTIONS: 5

Total Marks50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I D.HUSSAINAIH(192425300) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D.Hussainaiah

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of	15%	Clearly interprets	Basic	Limited or unclear	No interpretation	

Results		results with units and engineering relevance	interpretation with minor omissions	interpretation		
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement

Answer:

Grey level transformation is an image enhancement technique that changes the intensity values of pixels to improve image quality.

Purpose:

- Improves image brightness and contrast.
- Enhances important image details.
- Makes dark or bright regions more visible.
- Helps in better image analysis and processing.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

Given:

Formula:

$$s=255-r \quad s = 255 - r \quad s=255-r$$

Input pixel values:

$$r=[50, 100, 150, 200] \quad r=[50, \backslash; 100, \backslash; 150, \backslash; 200] \quad r=[50, 100, 150, 200]$$

Calculation:

For $r = 50$

$$s=255-50=205 \quad s=255-50=205 \quad s=255-50=205$$

For $r = 100$

$$s=255-100=155 \quad s=255-100=155 \quad s=255-100=155$$

For $r = 150$

$$s=255-150=105 \quad s=255-150=105 \quad s=255-150=105$$

For $r = 200$

$$s=255-200=55 \quad s=255-200=55 \quad s=255-200=55$$

Answer:

Input (r) Output (s)

50 205

100 155

150 105

200 55

Final Output:

[205, 155, 105, 55]\boxed{[205,\,;155,\,;105,\,;55]}[205,155,105,55]

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Answer:

Log transformation increases the intensity of dark pixels more than bright pixels.

Advantages:

- Enhances dark regions.
- Compresses high-intensity values.
- Improves visibility of details in low-light images.
- Widely used in medical and satellite image processing.

(b) Given $s = c \log_{10}(1+r)$, where $c=10$, compute s for:
 $r=[0,10,50,100]$

Given

$$s = 10 \log_{10}(1+r) \quad s = 10 \log(1+r) \quad s = 10 \log(1+r)$$

Using **log base 10**

Input:

$$r = [0, 10, 50, 100] \quad r = [0, \backslash; 10, \backslash; 50, \backslash; 100] \quad r = [0, 10, 50, 100]$$

Calculation

For **$r = 0$**

$$s = 10 \log_{10}(1) = 0 \quad s = 10 \log(1) = 0 \quad s = 10 \log(1) = 0$$

For **$r = 10$**

$$s = 10 \log_{10}(11) \quad s = 10 \log(11) \quad s = 10 \log(11) = 10(1.041) = 10.41 = 10(1.041) = 10.41 = 10(1.041) = 10.41$$

For **$r = 50$**

$$s = 10 \log_{10}(51) \quad s = 10 \log(51) \quad s = 10 \log(51) = 10(1.708) = 17.08 = 10(1.708) = 17.08 = 10(1.708) = 17.08$$

For $r = 100$

$$s = 10 \log_{f_0}(101) = 10 \log(101) = 10 \log(101) \\ = 10(2.004) = 20.04 = 10(2.004) = 20.04 = 10(2.004) = 20.04$$

Answer

r	s
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0	0
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10	10.41
----	-------

50	17.08
----	-------

100	20.04
-----	-------

Final Output:

[0, 10.41, 17.08, 20.04]
 $\boxed{[0, \backslash; 10.41, \backslash; 17.08, \backslash; 20.04]}$
[0,10.41,17.08,20.04]

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Answer:

Histogram equalization improves the contrast of an image by spreading pixel intensity values over the full range.

Advantages:

- Improves image contrast.
- Enhances hidden details.
- Produces a more uniform intensity distribution.
- Makes image features easier to detect.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Given Histogram

Intensity Frequency

0	2
1	2
2	4
3	8

Step 1: Total Pixels

$$N=2+2+4+8=16 \quad N=2+2+4+8=16 \quad N=2+2+4+8=16$$

Step 2: Cumulative Distribution

Intensity Frequency CDF

0	2	2
1	2	4
2	4	8
3	8	16

Step 3: Equalized Intensity

Formula

$$s = \text{CDF} \times (4-1) \quad s = \frac{\text{CDF}}{16} \times (4-1) \quad s = \frac{\text{CDF}}{16} \times 3$$

For intensity **0**

$$s = \frac{2}{16} \times 3 = 0.375 \approx 0 \quad s = \frac{2}{16} \times 3 = 0.375 \approx 0 \quad s = \frac{2}{16} \times 3 = 0.375 \approx 0$$

For intensity **1**

$$s = \frac{4}{16} \times 3 = 0.75 \approx 1 \quad s = \frac{4}{16} \times 3 = 0.75 \approx 1 \quad s = \frac{4}{16} \times 3 = 0.75 \approx 1$$

For intensity **2**

$$s = \frac{8}{16} \times 3 = 1.5 \approx 2 \quad s = \frac{8}{16} \times 3 = 1.5 \approx 2 \quad s = \frac{8}{16} \times 3 = 1.5 \approx 2$$

For intensity **3**

$$s = \frac{16}{16} \times 3 = 3 \quad s = \frac{16}{16} \times 3 = 3 \quad s = \frac{16}{16} \times 3 = 3$$

Final Table

Original Intensity Equalized Intensity

Original Intensity Equalized Intensity

0	0
1	1
2	2
3	3

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Answer:

Spatial smoothing reduces noise by replacing each pixel with the average value of its neighbouring pixels.

Purpose:

- Removes noise.
- Smooths the image.
- Reduces unwanted variations.
- Improves image quality before segmentation.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Given matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Step 1: Find the Sum

$$10+20+30+20+30+40+30+40+50=270$$

Step 2: Average

$$\frac{270}{9}=30$$

Answer

New centre pixel =

$$\boxed{30}$$

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Answer:

Sharpening highlights edges and fine details by increasing the difference between neighbouring pixel values.

Advantages:

- Enhances edges.
- Improves image clarity.
- Highlights fine details.
- Makes object boundaries more visible.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$
to the matrix:

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Compute the output at the center.

Laplacian Mask

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$
 $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Image Matrix

$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$
 $\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

Calculation

$$= (4 \times 20) - (10 + 10 + 10 + 10) = (4 \times 20) - (10 + 10 + 10 + 10) = 80 - 40 = 80 - 40 = 80 - 40 = 40 = 40 = 40$$

Answer

Output at the centre pixel =

$\boxed{40}$

Question

Answer

Q1(b) **[205, 155, 105, 55]**

Q2(b) **[0, 10.41, 17.08, 20.04]**

Q3(b) **CDF = [2, 4, 8, 16]; Equalized Intensities = [0, 1, 2, 3]**

Q4(b) **30**

Q5(b) **40**